Table 1: Threshold pilot only: eight simulation settings x one repetition per design, plus eight real datasets x two label budgets x one split per design in the table. All three rules apply to the same selected train-only model. Same-model quantile uses the population-representative validation quantile (all validation observations for one-sample; validation U only for two-sample). PU-estimated BA is a pilot rule; oracle BA uses true validation labels and is diagnostic only. Test labels never select thresholds. No historical mixed or refit-transfer rule is presented. The figure shows the six conceptual representatives on simulation cases; the table retains all 25 methods and real pilot cases. Six real-data method/case replays failed validation verification and are marked unavailable, leaving 1,194 verified method/case results per rule.

One-sample - real: Credit Default, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.671	0.658	0.671	yes
uPU logistic	0.674	0.661	0.672	yes
nnPU logistic	0.674	0.661	0.672	yes
Naive AdaBoost	0.691	0.649	0.691	yes
Naive GBM	0.676	0.654	0.676	yes
Naive XGBoost	0.675	0.671	0.681	yes
Naive LightGBM	0.669	0.680	0.684	yes
Naive CatBoost	0.682	0.661	0.683	yes
Corrected AdaBoost	0.691	0.649	0.691	yes
Corrected GBM	0.676	0.654	0.676	yes
Corrected XGBoost	0.675	0.671	0.681	yes
Corrected LightGBM	0.669	0.680	0.684	yes
Corrected CatBoost	0.682	0.661	0.683	yes
RN AdaBoost	0.642	0.682	0.682	yes
RN GBM	0.674	0.661	0.670	yes
RN XGBoost	0.673	0.672	0.669	yes
RN LightGBM	0.636	0.621	0.656	yes
RN CatBoost	0.667	0.645	0.667	yes
CS AdaBoost	0.691	0.649	0.691	yes
CS GBM	0.679	0.668	0.693	yes
CS XGBoost	0.678	0.675	0.677	yes
CS LightGBM	0.670	0.651	0.682	yes
CS CatBoost	0.681	0.669	0.691	yes
AdaPU	0.691	0.691	0.692	yes
PU gradient boosting	0.500	0.684	0.684	yes

One-sample - real: Credit Default, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.672	0.671	0.674	yes
uPU logistic	0.673	0.673	0.671	yes
nnPU logistic	0.673	0.673	0.671	yes
Naive AdaBoost	0.699	0.676	0.705	yes
Naive GBM	0.697	0.680	0.701	yes
Naive XGBoost	0.699	0.678	0.710	yes
Naive LightGBM	0.689	0.692	0.698	yes
Naive CatBoost	0.676	0.694	0.693	yes
Corrected AdaBoost	0.699	0.676	0.705	yes
Corrected GBM	0.697	0.680	0.701	yes
Corrected XGBoost	0.699	0.678	0.710	yes

One-sample - real: Credit Default, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Corrected LightGBM	0.689	0.692	0.698	yes
Corrected CatBoost	0.676	0.694	0.693	yes
RN AdaBoost	0.678	0.682	0.682	yes
RN GBM	0.636	0.650	0.650	yes
RN XGBoost	0.673	0.668	0.672	yes
RN LightGBM	0.672	0.641	0.672	yes
RN CatBoost	0.675	0.674	0.675	yes
CS AdaBoost	0.699	0.676	0.705	yes
CS GBM	0.691	0.704	0.707	yes
CS XGBoost	0.699	0.678	0.710	yes
CS LightGBM	0.693	0.705	0.705	yes
CS CatBoost	0.701	0.707	0.709	yes
AdaPU	0.700	0.704	0.705	yes
PU gradient boosting	0.702	0.672	0.702	yes

One-sample - real: Dry Bean, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.940	0.946	0.953	yes
uPU logistic	0.934	0.942	0.942	yes
nnPU logistic	0.934	0.942	0.942	yes
Naive AdaBoost	0.939	0.944	0.944	yes
Naive GBM	0.935	0.920	0.935	yes
Naive XGBoost	0.941	0.941	0.940	yes
Naive LightGBM	0.938	0.937	0.945	yes
Naive CatBoost	0.937	0.944	0.949	yes
Corrected AdaBoost	0.939	0.944	0.944	yes
Corrected GBM	0.935	0.920	0.935	yes
Corrected XGBoost	0.941	0.941	0.940	yes
Corrected LightGBM	0.938	0.937	0.945	yes
Corrected CatBoost	0.937	0.944	0.949	yes
RN AdaBoost	0.947	0.947	0.947	yes
RN GBM	0.944	0.947	0.947	yes
RN XGBoost	0.944	0.940	0.951	yes
RN LightGBM	0.946	0.939	0.951	yes
RN CatBoost	0.945	0.945	0.950	yes
CS AdaBoost	0.941	0.902	0.942	yes
CS GBM	0.937	0.860	0.939	yes
CS XGBoost	0.941	0.941	0.940	yes
CS LightGBM	0.940	0.934	0.943	yes
CS CatBoost	0.937	0.944	0.949	yes
AdaPU	0.910	0.889	0.920	yes
PU gradient boosting	0.891	0.910	0.915	yes

One-sample - real: Dry Bean, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.945	0.947	0.947	yes

One-sample - real: Dry Bean, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
uPU logistic	0.936	0.944	0.941	yes
nnPU logistic	0.936	0.944	0.941	yes
Naive AdaBoost	0.948	0.924	0.952	yes
Naive GBM	0.934	0.945	0.950	yes
Naive XGBoost	0.945	0.954	0.954	yes
Naive LightGBM	0.933	0.949	0.949	yes
Naive CatBoost	0.934	0.947	0.947	yes
Corrected AdaBoost	0.948	0.924	0.952	yes
Corrected GBM	0.934	0.945	0.950	yes
Corrected XGBoost	0.945	0.954	0.954	yes
Corrected LightGBM	0.933	0.949	0.949	yes
Corrected CatBoost	0.934	0.947	0.947	yes
RN AdaBoost	0.946	0.957	0.957	yes
RN GBM	0.948	0.952	0.951	yes
RN XGBoost	0.945	0.955	0.957	yes
RN LightGBM	0.947	0.956	0.957	yes
RN CatBoost	0.945	0.950	0.951	yes
CS AdaBoost	0.948	0.916	0.950	yes
CS GBM	0.940	0.924	0.949	yes
CS XGBoost	0.945	0.953	0.952	yes
CS LightGBM	0.932	0.950	0.951	yes
CS CatBoost	0.943	0.951	0.954	yes
AdaPU	0.888	0.902	0.905	yes
PU gradient boosting	0.944	0.939	0.946	yes

One-sample - real: HTRU2, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.916	0.905	0.927	yes
uPU logistic	0.925	0.936	0.933	yes
nnPU logistic	0.925	0.936	0.933	yes
Naive AdaBoost	0.913	0.887	0.921	yes
Naive GBM	0.916	0.894	0.909	yes
Naive XGBoost	0.916	0.919	0.919	yes
Naive LightGBM	0.901	0.885	0.917	yes
Naive CatBoost	0.917	0.897	0.929	yes
Corrected AdaBoost	0.913	0.887	0.921	yes
Corrected GBM	0.916	0.894	0.909	yes
Corrected XGBoost	0.916	0.919	0.919	yes
Corrected LightGBM	0.901	0.885	0.917	yes
Corrected CatBoost	0.917	0.897	0.929	yes
RN AdaBoost	0.922	0.898	0.922	yes
RN GBM	0.906	0.894	0.906	yes
RN XGBoost	0.912	0.900	0.914	yes
RN LightGBM	0.894	0.908	0.910	yes
RN CatBoost	0.916	0.900	0.928	yes
CS AdaBoost	0.910	0.899	0.910	yes
CS GBM	0.916	0.894	0.909	yes
CS XGBoost	0.898	0.894	0.906	yes

One-sample - real: HTRU2, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
CS LightGBM	0.873	0.899	0.895	yes
CS CatBoost	0.919	0.884	0.920	yes
AdaPU	0.889	0.884	0.889	yes
PU gradient boosting	0.912	0.915	0.915	yes

One-sample - real: HTRU2, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.905	0.918	0.917	yes
uPU logistic	0.915	0.925	0.925	yes
nnPU logistic	0.915	0.925	0.925	yes
Naive AdaBoost	0.917	0.914	0.922	yes
Naive GBM	0.918	0.927	0.930	yes
Naive XGBoost	0.921	0.924	0.922	yes
Naive LightGBM	0.908	0.922	0.922	yes
Naive CatBoost	0.918	0.931	0.930	yes
Corrected AdaBoost	0.917	0.914	0.922	yes
Corrected GBM	0.918	0.927	0.930	yes
Corrected XGBoost	0.921	0.924	0.922	yes
Corrected LightGBM	0.908	0.922	0.922	yes
Corrected CatBoost	0.918	0.931	0.930	yes
RN AdaBoost	0.922	0.919	0.922	yes
RN GBM	0.917	0.925	0.930	yes
RN XGBoost	0.919	0.923	0.931	yes
RN LightGBM	0.906	0.925	0.927	yes
RN CatBoost	0.902	0.929	0.928	yes
CS AdaBoost	0.917	0.914	0.922	yes
CS GBM	0.916	0.926	0.929	yes
CS XGBoost	0.921	0.924	0.922	yes
CS LightGBM	0.913	0.914	0.914	yes
CS CatBoost	0.922	0.924	0.921	yes
AdaPU	0.904	0.910	0.917	yes
PU gradient boosting	0.919	0.911	0.921	yes

One-sample - real: Online Shoppers, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.551	0.577	0.578	yes
uPU logistic	0.542	0.582	0.573	yes
nnPU logistic	0.550	0.580	0.568	yes
Naive AdaBoost	0.644	0.644	0.645	yes
Naive GBM	0.564	0.612	0.620	yes
Naive XGBoost	0.563	0.599	0.599	yes
Naive LightGBM	0.559	0.613	0.631	yes
Naive CatBoost	0.576	0.541	0.616	yes
Corrected AdaBoost	0.644	0.644	0.645	yes
Corrected GBM	0.564	0.612	0.620	yes
Corrected XGBoost	0.563	0.599	0.599	yes

One-sample - real: Online Shoppers, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Corrected LightGBM	0.559	0.613	0.631	yes
Corrected CatBoost	0.576	0.541	0.616	yes
RN AdaBoost	0.583	0.609	0.646	yes
RN GBM	0.551	0.568	0.618	yes
RN XGBoost	0.594	0.587	0.641	yes
RN LightGBM	0.557	0.617	0.613	yes
RN CatBoost	0.582	0.594	0.658	yes
CS AdaBoost	0.644	0.644	0.645	yes
CS GBM	0.539	0.559	0.592	yes
CS XGBoost	0.564	0.627	0.636	yes
CS LightGBM	0.567	0.598	0.635	yes
CS CatBoost	0.558	0.613	0.611	yes
AdaPU	0.597	0.634	0.629	yes
PU gradient boosting	0.515	0.629	0.629	yes

One-sample - real: Online Shoppers, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.580	0.637	0.637	yes
uPU logistic	0.583	0.630	0.630	yes
nnPU logistic	0.583	0.630	0.630	yes
Naive AdaBoost	0.628	0.656	0.657	yes
Naive GBM	0.616	0.661	0.670	yes
Naive XGBoost	0.590	0.645	0.659	yes
Naive LightGBM	0.592	0.639	0.636	yes
Naive CatBoost	0.596	0.658	0.664	yes
Corrected AdaBoost	0.628	0.656	0.657	yes
Corrected GBM	0.616	0.661	0.670	yes
Corrected XGBoost	0.590	0.645	0.659	yes
Corrected LightGBM	0.592	0.639	0.636	yes
Corrected CatBoost	0.596	0.658	0.664	yes
RN AdaBoost	0.624	0.644	0.640	yes
RN GBM	0.597	0.651	0.645	yes
RN XGBoost	0.611	0.652	0.647	yes
RN LightGBM	0.588	0.623	0.643	yes
RN CatBoost	0.588	0.653	0.653	yes
CS AdaBoost	0.604	0.653	0.653	yes
CS GBM	0.614	0.632	0.641	yes
CS XGBoost	0.590	0.645	0.659	yes
CS LightGBM	0.573	0.615	0.628	yes
CS CatBoost	0.588	0.643	0.651	yes
AdaPU	0.589	0.645	0.650	yes
PU gradient boosting	0.631	0.655	0.655	yes

One-sample - real: Optical Digits, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.770	0.777	0.776	yes

One-sample - real: Optical Digits, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
uPU logistic	0.761	0.741	0.761	yes
nnPU logistic	0.838	0.841	0.836	yes
Naive AdaBoost	0.816	0.755	0.812	yes
Naive GBM	0.732	0.721	0.732	yes
Naive XGBoost	0.762	0.721	0.764	yes
Naive LightGBM	0.788	0.773	0.788	yes
Naive CatBoost	0.840	0.834	0.837	yes
Corrected AdaBoost	0.816	0.755	0.812	yes
Corrected GBM	0.732	0.721	0.732	yes
Corrected XGBoost	0.762	0.721	0.764	yes
Corrected LightGBM	0.788	0.773	0.788	yes
Corrected CatBoost	0.840	0.834	0.837	yes
RN AdaBoost	–	–	–	no
RN GBM	0.827	0.811	0.820	yes
RN XGBoost	0.823	0.823	0.825	yes
RN LightGBM	0.819	0.738	0.826	yes
RN CatBoost	0.838	0.838	0.836	yes
CS AdaBoost	0.831	0.795	0.833	yes
CS GBM	0.744	0.744	0.740	yes
CS XGBoost	0.736	0.727	0.735	yes
CS LightGBM	0.793	0.770	0.794	yes
CS CatBoost	0.841	0.828	0.831	yes
AdaPU	0.802	0.782	0.802	yes
PU gradient boosting	0.774	0.778	0.780	yes

One-sample - real: Optical Digits, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.857	0.841	0.857	yes
uPU logistic	0.866	0.857	0.863	yes
nnPU logistic	0.866	0.855	0.859	yes
Naive AdaBoost	0.872	0.874	0.861	yes
Naive GBM	0.871	0.873	0.871	yes
Naive XGBoost	0.898	0.885	0.898	yes
Naive LightGBM	0.902	0.901	0.903	yes
Naive CatBoost	0.911	0.910	0.913	yes
Corrected AdaBoost	0.872	0.874	0.861	yes
Corrected GBM	0.871	0.873	0.871	yes
Corrected XGBoost	0.898	0.885	0.898	yes
Corrected LightGBM	0.902	0.901	0.903	yes
Corrected CatBoost	0.911	0.910	0.913	yes
RN AdaBoost	0.886	0.877	0.883	yes
RN GBM	0.864	0.859	0.867	yes
RN XGBoost	0.876	0.873	0.870	yes
RN LightGBM	0.898	0.888	0.889	yes
RN CatBoost	0.893	0.902	0.893	yes
CS AdaBoost	0.874	0.870	0.867	yes
CS GBM	0.890	0.890	0.890	yes
CS XGBoost	0.904	0.899	0.902	yes

One-sample - real: Optical Digits, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
CS LightGBM	0.902	0.895	0.903	yes
CS CatBoost	0.911	0.910	0.913	yes
AdaPU	0.849	0.862	0.833	yes
PU gradient boosting	0.875	0.868	0.870	yes

One-sample - real: Phishing Websites, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.875	0.867	0.876	yes
uPU logistic	0.818	0.753	0.818	yes
nnPU logistic	0.877	0.865	0.874	yes
Naive AdaBoost	0.896	0.896	0.900	yes
Naive GBM	0.875	0.883	0.881	yes
Naive XGBoost	0.867	0.819	0.849	yes
Naive LightGBM	0.866	0.854	0.867	yes
Naive CatBoost	0.858	0.852	0.846	yes
Corrected AdaBoost	0.896	0.896	0.900	yes
Corrected GBM	0.875	0.883	0.881	yes
Corrected XGBoost	0.867	0.819	0.849	yes
Corrected LightGBM	0.866	0.854	0.867	yes
Corrected CatBoost	0.858	0.852	0.846	yes
RN AdaBoost	0.888	0.888	0.925	yes
RN GBM	0.923	0.918	0.920	yes
RN XGBoost	0.914	0.903	0.910	yes
RN LightGBM	0.907	0.905	0.905	yes
RN CatBoost	0.911	0.908	0.904	yes
CS AdaBoost	0.898	0.851	0.915	yes
CS GBM	0.878	0.879	0.879	yes
CS XGBoost	0.835	0.827	0.828	yes
CS LightGBM	0.866	0.854	0.867	yes
CS CatBoost	0.858	0.852	0.846	yes
AdaPU	0.914	0.898	0.907	yes
PU gradient boosting	0.897	0.900	0.904	yes

One-sample - real: Phishing Websites, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.909	0.910	0.910	yes
uPU logistic	0.885	0.880	0.875	yes
nnPU logistic	0.889	0.887	0.883	yes
Naive AdaBoost	0.929	0.929	0.929	yes
Naive GBM	0.899	0.901	0.899	yes
Naive XGBoost	0.911	0.917	0.909	yes
Naive LightGBM	0.903	0.899	0.899	yes
Naive CatBoost	0.898	0.769	0.894	yes
Corrected AdaBoost	0.929	0.929	0.929	yes
Corrected GBM	0.899	0.901	0.899	yes
Corrected XGBoost	0.911	0.917	0.909	yes

One-sample - real: Phishing Websites, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Corrected LightGBM	0.903	0.899	0.899	yes
Corrected CatBoost	0.898	0.769	0.894	yes
RN AdaBoost	0.924	0.919	0.919	yes
RN GBM	0.921	0.886	0.926	yes
RN XGBoost	0.928	0.927	0.928	yes
RN LightGBM	0.924	0.900	0.922	yes
RN CatBoost	0.935	0.916	0.934	yes
CS AdaBoost	0.928	0.928	0.928	yes
CS GBM	0.899	0.901	0.899	yes
CS XGBoost	0.907	0.915	0.916	yes
CS LightGBM	0.891	0.898	0.891	yes
CS CatBoost	0.883	0.791	0.882	yes
AdaPU	0.916	0.914	0.914	yes
PU gradient boosting	0.915	0.896	0.920	yes

One-sample - real: Rice, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.904	0.880	0.902	yes
uPU logistic	0.892	0.855	0.901	yes
nnPU logistic	0.903	0.860	0.902	yes
Naive AdaBoost	0.887	0.811	0.878	yes
Naive GBM	0.864	0.690	0.861	yes
Naive XGBoost	0.869	0.801	0.885	yes
Naive LightGBM	0.844	0.667	0.889	yes
Naive CatBoost	0.846	0.706	0.847	yes
Corrected AdaBoost	0.887	0.811	0.878	yes
Corrected GBM	0.864	0.690	0.861	yes
Corrected XGBoost	0.869	0.801	0.885	yes
Corrected LightGBM	0.844	0.667	0.889	yes
Corrected CatBoost	0.846	0.706	0.847	yes
RN AdaBoost	0.918	0.827	0.910	yes
RN GBM	0.909	0.803	0.918	yes
RN XGBoost	0.889	0.649	0.919	yes
RN LightGBM	0.905	0.778	0.905	yes
RN CatBoost	0.912	0.829	0.921	yes
CS AdaBoost	0.886	0.810	0.875	yes
CS GBM	0.864	0.611	0.876	yes
CS XGBoost	0.807	0.642	0.823	yes
CS LightGBM	0.844	0.667	0.889	yes
CS CatBoost	0.846	0.706	0.847	yes
AdaPU	0.920	0.859	0.919	yes
PU gradient boosting	0.917	0.917	0.917	yes

One-sample - real: Rice, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.919	0.917	0.920	yes

One-sample - real: Rice, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
uPU logistic	0.924	0.923	0.924	yes
nnPU logistic	0.924	0.923	0.924	yes
Naive AdaBoost	0.906	0.884	0.912	yes
Naive GBM	0.897	0.915	0.891	yes
Naive XGBoost	0.902	0.926	0.902	yes
Naive LightGBM	0.880	0.899	0.904	yes
Naive CatBoost	0.905	0.897	0.905	yes
Corrected AdaBoost	0.906	0.884	0.912	yes
Corrected GBM	0.897	0.915	0.891	yes
Corrected XGBoost	0.902	0.926	0.902	yes
Corrected LightGBM	0.880	0.899	0.904	yes
Corrected CatBoost	0.905	0.897	0.905	yes
RN AdaBoost	0.901	0.895	0.922	yes
RN GBM	0.898	0.887	0.925	yes
RN XGBoost	0.885	0.900	0.907	yes
RN LightGBM	0.921	0.912	0.913	yes
RN CatBoost	0.921	0.891	0.925	yes
CS AdaBoost	0.906	0.884	0.912	yes
CS GBM	0.906	0.904	0.906	yes
CS XGBoost	0.891	0.881	0.906	yes
CS LightGBM	0.874	0.896	0.894	yes
CS CatBoost	0.905	0.897	0.905	yes
AdaPU	0.913	0.912	0.912	yes
PU gradient boosting	0.920	0.921	0.920	yes

One-sample - real: Spambase, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.793	0.774	0.794	yes
uPU logistic	0.775	0.689	0.768	yes
nnPU logistic	0.790	0.724	0.793	yes
Naive AdaBoost	0.813	0.656	0.821	yes
Naive GBM	0.807	0.721	0.807	yes
Naive XGBoost	0.802	0.661	0.794	yes
Naive LightGBM	0.701	0.696	0.690	yes
Naive CatBoost	0.867	0.651	0.872	yes
Corrected AdaBoost	0.813	0.656	0.821	yes
Corrected GBM	0.807	0.721	0.807	yes
Corrected XGBoost	0.802	0.661	0.794	yes
Corrected LightGBM	0.701	0.696	0.690	yes
Corrected CatBoost	0.867	0.651	0.872	yes
RN AdaBoost	–	–	–	no
RN GBM	0.824	0.769	0.840	yes
RN XGBoost	0.839	0.775	0.845	yes
RN LightGBM	0.827	0.660	0.830	yes
RN CatBoost	0.868	0.722	0.868	yes
CS AdaBoost	0.834	0.716	0.830	yes
CS GBM	0.795	0.731	0.787	yes
CS XGBoost	0.766	0.689	0.761	yes

One-sample - real: Spambase, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
CS LightGBM	0.759	0.757	0.751	yes
CS CatBoost	0.848	0.704	0.848	yes
AdaPU	0.835	0.698	0.842	yes
PU gradient boosting	0.795	0.744	0.789	yes

One-sample - real: Spambase, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.842	0.839	0.841	yes
uPU logistic	0.854	0.758	0.858	yes
nnPU logistic	0.864	0.783	0.871	yes
Naive AdaBoost	–	–	–	no
Naive GBM	0.882	0.888	0.876	yes
Naive XGBoost	0.891	0.857	0.897	yes
Naive LightGBM	0.881	0.866	0.883	yes
Naive CatBoost	0.912	0.897	0.908	yes
Corrected AdaBoost	–	–	–	no
Corrected GBM	0.882	0.888	0.876	yes
Corrected XGBoost	0.891	0.857	0.897	yes
Corrected LightGBM	0.881	0.866	0.883	yes
Corrected CatBoost	0.912	0.897	0.908	yes
RN AdaBoost	0.906	0.882	0.909	yes
RN GBM	0.917	0.917	0.910	yes
RN XGBoost	0.910	0.881	0.906	yes
RN LightGBM	0.909	0.888	0.908	yes
RN CatBoost	0.924	0.925	0.924	yes
CS AdaBoost	–	–	–	no
CS GBM	0.870	0.853	0.868	yes
CS XGBoost	0.869	0.841	0.871	yes
CS LightGBM	0.883	0.883	0.885	yes
CS CatBoost	0.912	0.897	0.908	yes
AdaPU	0.856	0.851	0.857	yes
PU gradient boosting	0.878	0.862	0.880	yes

Two-sample - real: Credit Default, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.655	0.657	0.662	yes
uPU logistic	0.661	0.650	0.666	yes
nnPU logistic	0.661	0.650	0.666	yes
Naive AdaBoost	0.637	0.675	0.674	yes
Naive GBM	0.676	0.685	0.682	yes
Naive XGBoost	0.672	0.671	0.674	yes
Naive LightGBM	0.654	0.658	0.669	yes
Naive CatBoost	0.686	0.695	0.689	yes
Corrected AdaBoost	0.637	0.675	0.674	yes
Corrected GBM	0.676	0.685	0.682	yes
Corrected XGBoost	0.672	0.671	0.674	yes

Two-sample - real: Credit Default, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Corrected LightGBM	0.654	0.658	0.669	yes
Corrected CatBoost	0.686	0.695	0.689	yes
RN AdaBoost	0.661	0.659	0.662	yes
RN GBM	0.670	0.670	0.660	yes
RN XGBoost	0.661	0.621	0.661	yes
RN LightGBM	0.594	0.594	0.627	yes
RN CatBoost	0.657	0.653	0.659	yes
CS AdaBoost	0.637	0.675	0.674	yes
CS GBM	0.676	0.685	0.682	yes
CS XGBoost	0.674	0.673	0.675	yes
CS LightGBM	0.654	0.658	0.669	yes
CS CatBoost	0.686	0.695	0.689	yes
AdaPU	0.693	0.684	0.689	yes
PU gradient boosting	0.680	0.684	0.689	yes

Two-sample - real: Credit Default, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.673	0.677	0.674	yes
uPU logistic	0.679	0.675	0.678	yes
nnPU logistic	0.679	0.675	0.678	yes
Naive AdaBoost	0.697	0.688	0.695	yes
Naive GBM	0.685	0.696	0.697	yes
Naive XGBoost	0.687	0.689	0.694	yes
Naive LightGBM	0.681	0.694	0.692	yes
Naive CatBoost	0.691	0.711	0.705	yes
Corrected AdaBoost	0.697	0.688	0.695	yes
Corrected GBM	0.685	0.696	0.697	yes
Corrected XGBoost	0.687	0.689	0.694	yes
Corrected LightGBM	0.681	0.694	0.692	yes
Corrected CatBoost	0.691	0.711	0.705	yes
RN AdaBoost	0.673	0.671	0.683	yes
RN GBM	0.673	0.667	0.676	yes
RN XGBoost	0.677	0.675	0.679	yes
RN LightGBM	0.645	0.651	0.652	yes
RN CatBoost	0.675	0.671	0.676	yes
CS AdaBoost	0.680	0.686	0.693	yes
CS GBM	0.685	0.697	0.696	yes
CS XGBoost	0.688	0.691	0.693	yes
CS LightGBM	0.681	0.694	0.692	yes
CS CatBoost	0.691	0.711	0.705	yes
AdaPU	0.691	0.696	0.694	yes
PU gradient boosting	0.689	0.685	0.689	yes

Two-sample - real: Dry Bean, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.946	0.947	0.948	yes

Two-sample - real: Dry Bean, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
uPU logistic	0.944	0.939	0.944	yes
nnPU logistic	0.944	0.939	0.944	yes
Naive AdaBoost	0.935	0.944	0.945	yes
Naive GBM	0.908	0.919	0.921	yes
Naive XGBoost	0.923	0.924	0.933	yes
Naive LightGBM	0.875	0.890	0.888	yes
Naive CatBoost	0.886	0.912	0.913	yes
Corrected AdaBoost	0.935	0.944	0.945	yes
Corrected GBM	0.908	0.919	0.921	yes
Corrected XGBoost	0.923	0.924	0.933	yes
Corrected LightGBM	0.875	0.890	0.888	yes
Corrected CatBoost	0.886	0.912	0.913	yes
RN AdaBoost	0.952	0.952	0.956	yes
RN GBM	0.951	0.917	0.950	yes
RN XGBoost	0.954	0.939	0.949	yes
RN LightGBM	0.948	0.955	0.955	yes
RN CatBoost	0.948	0.950	0.954	yes
CS AdaBoost	0.935	0.944	0.946	yes
CS GBM	0.889	0.900	0.900	yes
CS XGBoost	0.892	0.917	0.924	yes
CS LightGBM	0.875	0.890	0.888	yes
CS CatBoost	0.887	0.914	0.920	yes
AdaPU	0.898	0.901	0.909	yes
PU gradient boosting	0.942	0.926	0.944	yes

Two-sample - real: Dry Bean, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.942	0.944	0.946	yes
uPU logistic	0.934	0.937	0.938	yes
nnPU logistic	0.934	0.937	0.938	yes
Naive AdaBoost	0.942	0.948	0.947	yes
Naive GBM	0.934	0.946	0.944	yes
Naive XGBoost	0.935	0.947	0.946	yes
Naive LightGBM	0.921	0.951	0.950	yes
Naive CatBoost	0.951	0.949	0.949	yes
Corrected AdaBoost	0.942	0.948	0.947	yes
Corrected GBM	0.934	0.946	0.944	yes
Corrected XGBoost	0.935	0.947	0.946	yes
Corrected LightGBM	0.921	0.951	0.950	yes
Corrected CatBoost	0.951	0.949	0.949	yes
RN AdaBoost	0.946	0.945	0.952	yes
RN GBM	0.948	0.945	0.947	yes
RN XGBoost	0.943	0.949	0.948	yes
RN LightGBM	0.945	0.948	0.949	yes
RN CatBoost	0.940	0.943	0.951	yes
CS AdaBoost	0.937	0.948	0.947	yes
CS GBM	0.934	0.946	0.944	yes
CS XGBoost	0.938	0.945	0.946	yes

Two-sample - real: Dry Bean, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
CS LightGBM	0.921	0.951	0.950	yes
CS CatBoost	0.943	0.951	0.948	yes
AdaPU	0.892	0.908	0.911	yes
PU gradient boosting	0.940	0.943	0.948	yes

Two-sample - real: HTRU2, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.866	0.855	0.881	yes
uPU logistic	0.886	0.892	0.890	yes
nnPU logistic	0.886	0.892	0.890	yes
Naive AdaBoost	0.899	0.824	0.910	yes
Naive GBM	0.903	0.903	0.904	yes
Naive XGBoost	0.814	0.768	0.848	yes
Naive LightGBM	0.861	0.874	0.876	yes
Naive CatBoost	0.913	0.923	0.926	yes
Corrected AdaBoost	0.899	0.824	0.910	yes
Corrected GBM	0.903	0.903	0.904	yes
Corrected XGBoost	0.814	0.768	0.848	yes
Corrected LightGBM	0.861	0.874	0.876	yes
Corrected CatBoost	0.913	0.923	0.926	yes
RN AdaBoost	0.903	0.818	0.908	yes
RN GBM	0.907	0.807	0.908	yes
RN XGBoost	0.882	0.808	0.892	yes
RN LightGBM	0.849	0.850	0.879	yes
RN CatBoost	0.879	0.845	0.908	yes
CS AdaBoost	0.899	0.824	0.910	yes
CS GBM	0.908	0.781	0.907	yes
CS XGBoost	0.852	0.829	0.876	yes
CS LightGBM	0.866	0.889	0.891	yes
CS CatBoost	0.913	0.923	0.926	yes
AdaPU	0.899	0.887	0.907	yes
PU gradient boosting	0.897	0.904	0.912	yes

Two-sample - real: HTRU2, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.918	0.920	0.922	yes
uPU logistic	0.922	0.894	0.933	yes
nnPU logistic	0.922	0.894	0.933	yes
Naive AdaBoost	0.915	0.893	0.923	yes
Naive GBM	0.916	0.928	0.917	yes
Naive XGBoost	0.919	0.910	0.930	yes
Naive LightGBM	0.919	0.930	0.930	yes
Naive CatBoost	0.888	0.918	0.925	yes
Corrected AdaBoost	0.915	0.893	0.923	yes
Corrected GBM	0.916	0.928	0.917	yes
Corrected XGBoost	0.919	0.910	0.930	yes

Two-sample - real: HTRU2, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Corrected LightGBM	0.919	0.930	0.930	yes
Corrected CatBoost	0.888	0.918	0.925	yes
RN AdaBoost	0.914	0.897	0.922	yes
RN GBM	0.922	0.888	0.933	yes
RN XGBoost	0.922	0.912	0.927	yes
RN LightGBM	0.919	0.910	0.925	yes
RN CatBoost	0.919	0.913	0.929	yes
CS AdaBoost	0.915	0.893	0.924	yes
CS GBM	0.914	0.926	0.924	yes
CS XGBoost	0.919	0.910	0.930	yes
CS LightGBM	0.919	0.928	0.928	yes
CS CatBoost	0.888	0.918	0.925	yes
AdaPU	0.899	0.922	0.918	yes
PU gradient boosting	0.918	0.920	0.907	yes

Two-sample - real: Online Shoppers, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.539	0.523	0.551	yes
uPU logistic	0.538	0.524	0.560	yes
nnPU logistic	0.538	0.524	0.560	yes
Naive AdaBoost	0.581	0.652	0.610	yes
Naive GBM	0.563	0.601	0.599	yes
Naive XGBoost	0.559	0.631	0.617	yes
Naive LightGBM	0.556	0.633	0.611	yes
Naive CatBoost	0.569	0.607	0.657	yes
Corrected AdaBoost	0.581	0.652	0.610	yes
Corrected GBM	0.563	0.601	0.599	yes
Corrected XGBoost	0.559	0.631	0.617	yes
Corrected LightGBM	0.556	0.633	0.611	yes
Corrected CatBoost	0.569	0.607	0.657	yes
RN AdaBoost	0.593	0.632	0.663	yes
RN GBM	0.564	0.645	0.628	yes
RN XGBoost	0.583	0.642	0.627	yes
RN LightGBM	0.558	0.624	0.618	yes
RN CatBoost	0.561	0.643	0.645	yes
CS AdaBoost	0.581	0.652	0.610	yes
CS GBM	0.565	0.610	0.617	yes
CS XGBoost	0.559	0.631	0.617	yes
CS LightGBM	0.552	0.622	0.610	yes
CS CatBoost	0.569	0.607	0.657	yes
AdaPU	0.581	0.607	0.628	yes
PU gradient boosting	0.573	0.649	0.649	yes

Two-sample - real: Online Shoppers, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.581	0.621	0.654	yes

Two-sample - real: Online Shoppers, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
uPU logistic	0.585	0.633	0.615	yes
nnPU logistic	0.585	0.633	0.615	yes
Naive AdaBoost	0.595	0.667	0.669	yes
Naive GBM	0.587	0.615	0.660	yes
Naive XGBoost	0.590	0.647	0.655	yes
Naive LightGBM	0.572	0.618	0.666	yes
Naive CatBoost	0.615	0.678	0.683	yes
Corrected AdaBoost	0.595	0.667	0.669	yes
Corrected GBM	0.587	0.615	0.660	yes
Corrected XGBoost	0.590	0.647	0.655	yes
Corrected LightGBM	0.572	0.618	0.666	yes
Corrected CatBoost	0.615	0.678	0.683	yes
RN AdaBoost	0.617	0.640	0.637	yes
RN GBM	0.584	0.658	0.646	yes
RN XGBoost	0.597	0.643	0.642	yes
RN LightGBM	0.577	0.621	0.640	yes
RN CatBoost	0.579	0.653	0.640	yes
CS AdaBoost	0.596	0.667	0.669	yes
CS GBM	0.595	0.640	0.667	yes
CS XGBoost	0.590	0.647	0.655	yes
CS LightGBM	0.572	0.618	0.666	yes
CS CatBoost	0.615	0.678	0.683	yes
AdaPU	0.587	0.618	0.659	yes
PU gradient boosting	0.600	0.674	0.670	yes

Two-sample - real: Optical Digits, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.764	0.759	0.761	yes
uPU logistic	0.778	0.737	0.775	yes
nnPU logistic	0.849	0.855	0.847	yes
Naive AdaBoost	0.761	0.760	0.738	yes
Naive GBM	0.720	0.716	0.714	yes
Naive XGBoost	0.731	0.667	0.733	yes
Naive LightGBM	0.782	0.746	0.783	yes
Naive CatBoost	0.820	0.786	0.816	yes
Corrected AdaBoost	0.761	0.760	0.738	yes
Corrected GBM	0.720	0.716	0.714	yes
Corrected XGBoost	0.731	0.667	0.733	yes
Corrected LightGBM	0.782	0.746	0.783	yes
Corrected CatBoost	0.820	0.786	0.816	yes
RN AdaBoost	–	–	–	no
RN GBM	0.799	0.714	0.797	yes
RN XGBoost	0.823	0.781	0.825	yes
RN LightGBM	0.805	0.726	0.797	yes
RN CatBoost	0.879	0.838	0.878	yes
CS AdaBoost	0.761	0.760	0.738	yes
CS GBM	0.720	0.716	0.714	yes
CS XGBoost	0.770	0.768	0.771	yes

Two-sample - real: Optical Digits, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
CS LightGBM	0.768	0.758	0.771	yes
CS CatBoost	0.818	0.749	0.819	yes
AdaPU	0.722	0.706	0.719	yes
PU gradient boosting	0.647	0.677	0.680	yes

Two-sample - real: Optical Digits, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.855	0.854	0.854	yes
uPU logistic	0.810	0.810	0.812	yes
nnPU logistic	0.874	0.870	0.874	yes
Naive AdaBoost	0.875	0.838	0.872	yes
Naive GBM	0.875	0.794	0.873	yes
Naive XGBoost	0.904	0.853	0.901	yes
Naive LightGBM	0.896	0.843	0.896	yes
Naive CatBoost	0.899	0.867	0.901	yes
Corrected AdaBoost	0.875	0.838	0.872	yes
Corrected GBM	0.875	0.794	0.873	yes
Corrected XGBoost	0.904	0.853	0.901	yes
Corrected LightGBM	0.896	0.843	0.896	yes
Corrected CatBoost	0.899	0.867	0.901	yes
RN AdaBoost	0.885	0.867	0.887	yes
RN GBM	0.903	0.864	0.895	yes
RN XGBoost	0.908	0.882	0.907	yes
RN LightGBM	0.908	0.892	0.907	yes
RN CatBoost	0.908	0.874	0.901	yes
CS AdaBoost	0.872	0.832	0.866	yes
CS GBM	0.894	0.886	0.890	yes
CS XGBoost	0.903	0.881	0.898	yes
CS LightGBM	0.899	0.828	0.899	yes
CS CatBoost	0.915	0.840	0.913	yes
AdaPU	0.850	0.848	0.849	yes
PU gradient boosting	0.838	0.789	0.851	yes

Two-sample - real: Phishing Websites, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.793	0.792	0.792	yes
uPU logistic	0.767	0.754	0.771	yes
nnPU logistic	0.811	0.804	0.814	yes
Naive AdaBoost	0.864	0.864	0.864	yes
Naive GBM	0.832	0.793	0.830	yes
Naive XGBoost	0.832	0.799	0.830	yes
Naive LightGBM	0.818	0.818	0.826	yes
Naive CatBoost	0.850	0.846	0.851	yes
Corrected AdaBoost	0.864	0.864	0.864	yes
Corrected GBM	0.832	0.793	0.830	yes
Corrected XGBoost	0.832	0.799	0.830	yes

Two-sample - real: Phishing Websites, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Corrected LightGBM	0.818	0.818	0.826	yes
Corrected CatBoost	0.850	0.846	0.851	yes
RN AdaBoost	0.853	0.884	0.887	yes
RN GBM	0.836	0.840	0.832	yes
RN XGBoost	0.831	0.832	0.827	yes
RN LightGBM	0.823	0.827	0.825	yes
RN CatBoost	0.854	0.847	0.846	yes
CS AdaBoost	0.866	0.866	0.866	yes
CS GBM	0.818	0.794	0.818	yes
CS XGBoost	0.830	0.818	0.827	yes
CS LightGBM	0.826	0.819	0.829	yes
CS CatBoost	0.851	0.850	0.854	yes
AdaPU	0.899	0.893	0.899	yes
PU gradient boosting	0.500	0.887	0.887	yes

Two-sample - real: Phishing Websites, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.918	0.918	0.919	yes
uPU logistic	0.879	0.878	0.875	yes
nnPU logistic	0.924	0.899	0.921	yes
Naive AdaBoost	0.923	0.930	0.930	yes
Naive GBM	0.933	0.933	0.932	yes
Naive XGBoost	0.926	0.922	0.924	yes
Naive LightGBM	0.926	0.926	0.921	yes
Naive CatBoost	0.921	0.923	0.928	yes
Corrected AdaBoost	0.923	0.930	0.930	yes
Corrected GBM	0.933	0.933	0.932	yes
Corrected XGBoost	0.926	0.922	0.924	yes
Corrected LightGBM	0.926	0.926	0.921	yes
Corrected CatBoost	0.921	0.923	0.928	yes
RN AdaBoost	0.927	0.869	0.921	yes
RN GBM	0.926	0.787	0.929	yes
RN XGBoost	0.931	0.932	0.921	yes
RN LightGBM	0.927	0.890	0.929	yes
RN CatBoost	0.929	0.928	0.931	yes
CS AdaBoost	0.923	0.930	0.930	yes
CS GBM	0.929	0.929	0.931	yes
CS XGBoost	0.930	0.927	0.931	yes
CS LightGBM	0.926	0.926	0.921	yes
CS CatBoost	0.923	0.926	0.926	yes
AdaPU	0.916	0.838	0.912	yes
PU gradient boosting	0.902	0.908	0.908	yes

Two-sample - real: Rice, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.926	0.785	0.925	yes

Two-sample - real: Rice, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
uPU logistic	0.898	0.833	0.902	yes
nnPU logistic	0.903	0.829	0.901	yes
Naive AdaBoost	0.874	0.902	0.874	yes
Naive GBM	0.864	0.834	0.904	yes
Naive XGBoost	0.861	0.860	0.861	yes
Naive LightGBM	0.847	0.756	0.836	yes
Naive CatBoost	0.898	0.790	0.897	yes
Corrected AdaBoost	0.874	0.902	0.874	yes
Corrected GBM	0.864	0.834	0.904	yes
Corrected XGBoost	0.861	0.860	0.861	yes
Corrected LightGBM	0.847	0.756	0.836	yes
Corrected CatBoost	0.898	0.790	0.897	yes
RN AdaBoost	0.855	0.783	0.898	yes
RN GBM	0.860	0.728	0.896	yes
RN XGBoost	0.875	0.797	0.890	yes
RN LightGBM	0.872	0.785	0.880	yes
RN CatBoost	0.895	0.798	0.908	yes
CS AdaBoost	0.874	0.902	0.874	yes
CS GBM	0.869	0.848	0.854	yes
CS XGBoost	0.861	0.860	0.861	yes
CS LightGBM	0.856	0.855	0.847	yes
CS CatBoost	0.903	0.799	0.902	yes
AdaPU	0.876	0.806	0.901	yes
PU gradient boosting	0.861	0.810	0.906	yes

Two-sample - real: Rice, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.920	0.931	0.921	yes
uPU logistic	0.924	0.873	0.926	yes
nnPU logistic	0.924	0.873	0.926	yes
Naive AdaBoost	0.907	0.918	0.923	yes
Naive GBM	0.891	0.892	0.887	yes
Naive XGBoost	0.877	0.894	0.892	yes
Naive LightGBM	0.870	0.877	0.897	yes
Naive CatBoost	0.839	0.878	0.836	yes
Corrected AdaBoost	0.907	0.918	0.923	yes
Corrected GBM	0.891	0.892	0.887	yes
Corrected XGBoost	0.877	0.894	0.892	yes
Corrected LightGBM	0.870	0.877	0.897	yes
Corrected CatBoost	0.839	0.878	0.836	yes
RN AdaBoost	0.921	0.907	0.923	yes
RN GBM	0.923	0.894	0.922	yes
RN XGBoost	0.908	0.901	0.924	yes
RN LightGBM	0.898	0.863	0.918	yes
RN CatBoost	0.910	0.920	0.912	yes
CS AdaBoost	0.907	0.918	0.923	yes
CS GBM	0.868	0.865	0.877	yes
CS XGBoost	0.877	0.894	0.892	yes

Two-sample - real: Rice, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
CS LightGBM	0.870	0.877	0.897	yes
CS CatBoost	0.839	0.878	0.836	yes
AdaPU	0.900	0.900	0.897	yes
PU gradient boosting	0.926	0.904	0.925	yes

Two-sample - real: Spambase, budget 0.05

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.803	0.702	0.803	yes
uPU logistic	0.847	0.845	0.848	yes
nnPU logistic	0.847	0.845	0.848	yes
Naive AdaBoost	0.817	0.806	0.838	yes
Naive GBM	0.796	0.801	0.809	yes
Naive XGBoost	0.807	0.802	0.804	yes
Naive LightGBM	0.785	0.796	0.775	yes
Naive CatBoost	0.842	0.862	0.860	yes
Corrected AdaBoost	0.817	0.806	0.838	yes
Corrected GBM	0.796	0.801	0.809	yes
Corrected XGBoost	0.807	0.802	0.804	yes
Corrected LightGBM	0.785	0.796	0.775	yes
Corrected CatBoost	0.842	0.862	0.860	yes
RN AdaBoost	0.870	0.770	0.855	yes
RN GBM	0.869	0.892	0.890	yes
RN XGBoost	0.880	0.825	0.869	yes
RN LightGBM	0.862	0.792	0.855	yes
RN CatBoost	0.892	0.844	0.898	yes
CS AdaBoost	0.823	0.767	0.825	yes
CS GBM	0.806	0.772	0.803	yes
CS XGBoost	0.833	0.813	0.833	yes
CS LightGBM	0.785	0.796	0.775	yes
CS CatBoost	0.847	0.855	0.862	yes
AdaPU	0.864	0.866	0.870	yes
PU gradient boosting	0.791	0.816	0.778	yes

Two-sample - real: Spambase, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.844	0.847	0.848	yes
uPU logistic	0.869	0.879	0.882	yes
nnPU logistic	0.869	0.879	0.882	yes
Naive AdaBoost	0.886	0.883	0.885	yes
Naive GBM	0.844	0.792	0.851	yes
Naive XGBoost	0.841	0.795	0.840	yes
Naive LightGBM	0.854	0.822	0.859	yes
Naive CatBoost	0.853	0.871	0.854	yes
Corrected AdaBoost	0.886	0.883	0.885	yes
Corrected GBM	0.844	0.792	0.851	yes
Corrected XGBoost	0.841	0.795	0.840	yes

Two-sample - real: Spambase, budget 0.20

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Corrected LightGBM	0.854	0.822	0.859	yes
Corrected CatBoost	0.853	0.871	0.854	yes
RN AdaBoost	0.914	0.903	0.908	yes
RN GBM	0.927	0.927	0.927	yes
RN XGBoost	0.914	0.903	0.909	yes
RN LightGBM	0.909	0.903	0.907	yes
RN CatBoost	0.931	0.921	0.920	yes
CS AdaBoost	0.886	0.869	0.892	yes
CS GBM	0.832	0.835	0.830	yes
CS XGBoost	0.841	0.795	0.840	yes
CS LightGBM	0.823	0.808	0.821	yes
CS CatBoost	0.853	0.871	0.854	yes
AdaPU	0.881	0.884	0.884	yes
PU gradient boosting	0.896	0.894	0.893	yes

One-sample - simulation: Reference

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.630	0.657	0.657	yes
uPU logistic	0.630	0.657	0.656	yes
nnPU logistic	0.630	0.657	0.656	yes
Naive AdaBoost	0.622	0.639	0.640	yes
Naive GBM	0.620	0.648	0.647	yes
Naive XGBoost	0.621	0.643	0.643	yes
Naive LightGBM	0.609	0.630	0.634	yes
Naive CatBoost	0.625	0.655	0.652	yes
Corrected AdaBoost	0.622	0.639	0.640	yes
Corrected GBM	0.620	0.648	0.647	yes
Corrected XGBoost	0.621	0.643	0.643	yes
Corrected LightGBM	0.609	0.630	0.634	yes
Corrected CatBoost	0.625	0.655	0.652	yes
RN AdaBoost	0.615	0.644	0.644	yes
RN GBM	0.614	0.651	0.653	yes
RN XGBoost	0.603	0.646	0.648	yes
RN LightGBM	0.613	0.648	0.647	yes
RN CatBoost	0.615	0.652	0.649	yes
CS AdaBoost	0.620	0.637	0.639	yes
CS GBM	0.622	0.632	0.633	yes
CS XGBoost	0.619	0.649	0.649	yes
CS LightGBM	0.605	0.627	0.626	yes
CS CatBoost	0.614	0.639	0.640	yes
AdaPU	0.628	0.649	0.649	yes
PU gradient boosting	0.601	0.638	0.639	yes

One-sample - simulation: Prevalence $\pi = 0.01$

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.503	0.553	0.561	yes

One-sample - simulation: Prevalence $\pi = 0.01$

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
uPU logistic	0.503	0.565	0.562	yes
nnPU logistic	0.503	0.565	0.562	yes
Naive AdaBoost	0.507	0.577	0.577	yes
Naive GBM	0.506	0.505	0.512	yes
Naive XGBoost	0.507	0.543	0.571	yes
Naive LightGBM	0.499	0.552	0.597	yes
Naive CatBoost	0.512	0.586	0.580	yes
Corrected AdaBoost	0.507	0.577	0.577	yes
Corrected GBM	0.506	0.505	0.512	yes
Corrected XGBoost	0.507	0.543	0.571	yes
Corrected LightGBM	0.499	0.552	0.597	yes
Corrected CatBoost	0.512	0.586	0.580	yes
RN AdaBoost	0.493	0.545	0.545	yes
RN GBM	0.497	0.594	0.558	yes
RN XGBoost	0.512	0.578	0.618	yes
RN LightGBM	0.512	0.590	0.552	yes
RN CatBoost	0.503	0.549	0.607	yes
CS AdaBoost	0.526	0.568	0.553	yes
CS GBM	0.503	0.545	0.549	yes
CS XGBoost	0.507	0.590	0.543	yes
CS LightGBM	0.511	0.579	0.530	yes
CS CatBoost	0.512	0.586	0.580	yes
AdaPU	0.495	0.526	0.599	yes
PU gradient boosting	0.496	0.498	0.498	yes

One-sample - simulation: Prevalence $\pi = 0.05$

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.564	0.641	0.634	yes
uPU logistic	0.564	0.637	0.636	yes
nnPU logistic	0.564	0.637	0.636	yes
Naive AdaBoost	0.540	0.634	0.627	yes
Naive GBM	0.538	0.611	0.640	yes
Naive XGBoost	0.550	0.604	0.643	yes
Naive LightGBM	0.540	0.629	0.630	yes
Naive CatBoost	0.548	0.645	0.645	yes
Corrected AdaBoost	0.540	0.634	0.627	yes
Corrected GBM	0.538	0.611	0.640	yes
Corrected XGBoost	0.550	0.604	0.643	yes
Corrected LightGBM	0.540	0.629	0.630	yes
Corrected CatBoost	0.548	0.645	0.645	yes
RN AdaBoost	0.546	0.633	0.636	yes
RN GBM	0.551	0.650	0.650	yes
RN XGBoost	0.544	0.644	0.641	yes
RN LightGBM	0.549	0.621	0.622	yes
RN CatBoost	0.550	0.650	0.646	yes
CS AdaBoost	0.543	0.629	0.629	yes
CS GBM	0.531	0.633	0.627	yes
CS XGBoost	0.546	0.633	0.628	yes

One-sample - simulation: Prevalence $\pi = 0.05$

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
CS LightGBM	0.545	0.616	0.623	yes
CS CatBoost	0.548	0.645	0.645	yes
AdaPU	0.531	0.619	0.627	yes
PU gradient boosting	0.518	0.574	0.591	yes

One-sample - simulation: Prevalence $\pi = 0.50$

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.674	0.673	0.672	yes
uPU logistic	0.673	0.673	0.673	yes
nnPU logistic	0.673	0.673	0.673	yes
Naive AdaBoost	0.655	0.648	0.655	yes
Naive GBM	0.660	0.651	0.659	yes
Naive XGBoost	0.659	0.657	0.661	yes
Naive LightGBM	0.655	0.657	0.658	yes
Naive CatBoost	0.663	0.660	0.664	yes
Corrected AdaBoost	0.655	0.648	0.655	yes
Corrected GBM	0.660	0.651	0.659	yes
Corrected XGBoost	0.659	0.657	0.661	yes
Corrected LightGBM	0.655	0.657	0.658	yes
Corrected CatBoost	0.663	0.660	0.664	yes
RN AdaBoost	0.665	0.667	0.666	yes
RN GBM	0.661	0.659	0.661	yes
RN XGBoost	0.667	0.665	0.665	yes
RN LightGBM	0.668	0.665	0.666	yes
RN CatBoost	0.670	0.663	0.670	yes
CS AdaBoost	0.661	0.648	0.659	yes
CS GBM	0.659	0.642	0.661	yes
CS XGBoost	0.661	0.661	0.657	yes
CS LightGBM	0.655	0.657	0.658	yes
CS CatBoost	0.663	0.660	0.664	yes
AdaPU	0.663	0.655	0.661	yes
PU gradient boosting	0.654	0.638	0.655	yes

One-sample - simulation: High dimension ($p = 500$)

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.574	0.589	0.590	yes
uPU logistic	0.566	0.577	0.577	yes
nnPU logistic	0.566	0.577	0.577	yes
Naive AdaBoost	0.636	0.659	0.657	yes
Naive GBM	0.625	0.644	0.647	yes
Naive XGBoost	0.624	0.649	0.652	yes
Naive LightGBM	0.618	0.640	0.643	yes
Naive CatBoost	0.630	0.652	0.652	yes
Corrected AdaBoost	0.636	0.659	0.657	yes
Corrected GBM	0.625	0.644	0.647	yes
Corrected XGBoost	0.624	0.649	0.652	yes

One-sample - simulation: High dimension ($p = 500$)

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Corrected LightGBM	0.618	0.640	0.643	yes
Corrected CatBoost	0.630	0.652	0.652	yes
RN AdaBoost	0.642	0.662	0.668	yes
RN GBM	0.636	0.660	0.661	yes
RN XGBoost	0.633	0.661	0.663	yes
RN LightGBM	0.632	0.660	0.660	yes
RN CatBoost	0.632	0.647	0.649	yes
CS AdaBoost	0.638	0.658	0.662	yes
CS GBM	0.631	0.639	0.649	yes
CS XGBoost	0.632	0.652	0.656	yes
CS LightGBM	0.620	0.640	0.640	yes
CS CatBoost	0.630	0.652	0.652	yes
AdaPU	0.631	0.659	0.651	yes
PU gradient boosting	0.618	0.644	0.649	yes

One-sample - simulation: Curved boundary

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.609	0.628	0.627	yes
uPU logistic	0.609	0.628	0.628	yes
nnPU logistic	0.609	0.628	0.628	yes
Naive AdaBoost	0.606	0.637	0.636	yes
Naive GBM	0.601	0.633	0.631	yes
Naive XGBoost	0.610	0.639	0.635	yes
Naive LightGBM	0.605	0.619	0.624	yes
Naive CatBoost	0.606	0.635	0.633	yes
Corrected AdaBoost	0.606	0.637	0.636	yes
Corrected GBM	0.601	0.633	0.631	yes
Corrected XGBoost	0.610	0.639	0.635	yes
Corrected LightGBM	0.605	0.619	0.624	yes
Corrected CatBoost	0.606	0.635	0.633	yes
RN AdaBoost	0.599	0.632	0.631	yes
RN GBM	0.593	0.632	0.632	yes
RN XGBoost	0.602	0.630	0.630	yes
RN LightGBM	0.597	0.627	0.627	yes
RN CatBoost	0.602	0.621	0.625	yes
CS AdaBoost	0.609	0.639	0.639	yes
CS GBM	0.604	0.633	0.635	yes
CS XGBoost	0.607	0.636	0.636	yes
CS LightGBM	0.604	0.628	0.628	yes
CS CatBoost	0.601	0.632	0.638	yes
AdaPU	0.601	0.613	0.623	yes
PU gradient boosting	0.607	0.626	0.620	yes

One-sample - simulation: Feature interaction

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.582	0.606	0.605	yes

One-sample - simulation: Feature interaction

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
uPU logistic	0.582	0.606	0.605	yes
nnPU logistic	0.582	0.606	0.605	yes
Naive AdaBoost	0.618	0.619	0.619	yes
Naive GBM	0.589	0.615	0.614	yes
Naive XGBoost	0.592	0.613	0.614	yes
Naive LightGBM	0.594	0.625	0.626	yes
Naive CatBoost	0.616	0.640	0.640	yes
Corrected AdaBoost	0.618	0.619	0.619	yes
Corrected GBM	0.589	0.615	0.614	yes
Corrected XGBoost	0.592	0.613	0.614	yes
Corrected LightGBM	0.594	0.625	0.626	yes
Corrected CatBoost	0.616	0.640	0.640	yes
RN AdaBoost	0.537	0.607	0.612	yes
RN GBM	0.548	0.602	0.600	yes
RN XGBoost	0.554	0.604	0.606	yes
RN LightGBM	0.551	0.615	0.615	yes
RN CatBoost	0.552	0.598	0.597	yes
CS AdaBoost	0.618	0.619	0.619	yes
CS GBM	0.592	0.622	0.622	yes
CS XGBoost	0.598	0.623	0.619	yes
CS LightGBM	0.595	0.606	0.622	yes
CS CatBoost	0.628	0.626	0.651	yes
AdaPU	0.580	0.605	0.608	yes
PU gradient boosting	0.591	0.628	0.626	yes

One-sample - simulation: Unequal class covariance

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.637	0.646	0.646	yes
uPU logistic	0.637	0.646	0.644	yes
nnPU logistic	0.637	0.646	0.644	yes
Naive AdaBoost	0.653	0.663	0.669	yes
Naive GBM	0.654	0.659	0.663	yes
Naive XGBoost	0.648	0.657	0.658	yes
Naive LightGBM	0.638	0.642	0.650	yes
Naive CatBoost	0.659	0.674	0.670	yes
Corrected AdaBoost	0.653	0.663	0.669	yes
Corrected GBM	0.654	0.659	0.663	yes
Corrected XGBoost	0.648	0.657	0.658	yes
Corrected LightGBM	0.638	0.642	0.650	yes
Corrected CatBoost	0.659	0.674	0.670	yes
RN AdaBoost	0.640	0.641	0.645	yes
RN GBM	0.629	0.629	0.638	yes
RN XGBoost	0.623	0.634	0.634	yes
RN LightGBM	0.609	0.626	0.625	yes
RN CatBoost	0.627	0.630	0.631	yes
CS AdaBoost	0.651	0.667	0.668	yes
CS GBM	0.652	0.666	0.666	yes
CS XGBoost	0.649	0.656	0.654	yes

One-sample - simulation: Unequal class covariance

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
CS LightGBM	0.638	0.642	0.650	yes
CS CatBoost	0.659	0.666	0.666	yes
AdaPU	0.667	0.677	0.676	yes
PU gradient boosting	0.642	0.658	0.658	yes

Two-sample - simulation: Reference

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.643	0.660	0.670	yes
uPU logistic	0.642	0.663	0.669	yes
nnPU logistic	0.642	0.663	0.669	yes
Naive AdaBoost	0.637	0.658	0.652	yes
Naive GBM	0.632	0.659	0.660	yes
Naive XGBoost	0.632	0.647	0.660	yes
Naive LightGBM	0.629	0.648	0.654	yes
Naive CatBoost	0.634	0.658	0.659	yes
Corrected AdaBoost	0.637	0.658	0.652	yes
Corrected GBM	0.632	0.659	0.660	yes
Corrected XGBoost	0.632	0.647	0.660	yes
Corrected LightGBM	0.629	0.648	0.654	yes
Corrected CatBoost	0.634	0.658	0.659	yes
RN AdaBoost	0.634	0.667	0.666	yes
RN GBM	0.635	0.669	0.669	yes
RN XGBoost	0.634	0.667	0.669	yes
RN LightGBM	0.631	0.670	0.668	yes
RN CatBoost	0.636	0.669	0.671	yes
CS AdaBoost	0.637	0.652	0.658	yes
CS GBM	0.632	0.659	0.660	yes
CS XGBoost	0.632	0.647	0.660	yes
CS LightGBM	0.629	0.648	0.654	yes
CS CatBoost	0.634	0.658	0.659	yes
AdaPU	0.631	0.644	0.648	yes
PU gradient boosting	0.635	0.651	0.651	yes

Two-sample - simulation: Prevalence $\pi = 0.01$

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.556	0.681	0.683	yes
uPU logistic	0.557	0.670	0.670	yes
nnPU logistic	0.557	0.670	0.670	yes
Naive AdaBoost	0.556	0.688	0.688	yes
Naive GBM	0.568	0.671	0.690	yes
Naive XGBoost	0.557	0.667	0.690	yes
Naive LightGBM	0.545	0.648	0.639	yes
Naive CatBoost	0.551	0.677	0.694	yes
Corrected AdaBoost	0.556	0.688	0.688	yes
Corrected GBM	0.568	0.671	0.690	yes
Corrected XGBoost	0.557	0.667	0.690	yes

Two-sample - simulation: Prevalence $\pi = 0.01$

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Corrected LightGBM	0.545	0.648	0.639	yes
Corrected CatBoost	0.551	0.677	0.694	yes
RN AdaBoost	0.538	0.686	0.667	yes
RN GBM	0.533	0.669	0.663	yes
RN XGBoost	0.528	0.678	0.674	yes
RN LightGBM	0.556	0.677	0.663	yes
RN CatBoost	0.527	0.669	0.670	yes
CS AdaBoost	0.561	0.690	0.688	yes
CS GBM	0.568	0.671	0.690	yes
CS XGBoost	0.574	0.672	0.686	yes
CS LightGBM	0.545	0.648	0.639	yes
CS CatBoost	0.551	0.677	0.694	yes
AdaPU	0.545	0.666	0.641	yes
PU gradient boosting	0.550	0.654	0.654	yes

Two-sample - simulation: Prevalence $\pi = 0.05$

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.569	0.674	0.678	yes
uPU logistic	0.570	0.675	0.673	yes
nnPU logistic	0.570	0.675	0.673	yes
Naive AdaBoost	0.566	0.651	0.649	yes
Naive GBM	0.563	0.657	0.659	yes
Naive XGBoost	0.571	0.655	0.660	yes
Naive LightGBM	0.576	0.658	0.654	yes
Naive CatBoost	0.564	0.651	0.658	yes
Corrected AdaBoost	0.566	0.651	0.649	yes
Corrected GBM	0.563	0.657	0.659	yes
Corrected XGBoost	0.571	0.655	0.660	yes
Corrected LightGBM	0.576	0.658	0.654	yes
Corrected CatBoost	0.564	0.651	0.658	yes
RN AdaBoost	0.569	0.668	0.659	yes
RN GBM	0.569	0.660	0.663	yes
RN XGBoost	0.564	0.673	0.669	yes
RN LightGBM	0.572	0.676	0.665	yes
RN CatBoost	0.571	0.677	0.667	yes
CS AdaBoost	0.566	0.651	0.649	yes
CS GBM	0.563	0.657	0.659	yes
CS XGBoost	0.569	0.661	0.660	yes
CS LightGBM	0.576	0.658	0.654	yes
CS CatBoost	0.564	0.651	0.658	yes
AdaPU	0.566	0.650	0.649	yes
PU gradient boosting	0.558	0.633	0.633	yes

Two-sample - simulation: Prevalence $\pi = 0.50$

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.647	0.645	0.648	yes

Two-sample - simulation: Prevalence $\pi = 0.50$

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
uPU logistic	0.646	0.644	0.644	yes
nnPU logistic	0.646	0.644	0.644	yes
Naive AdaBoost	0.636	0.641	0.644	yes
Naive GBM	0.648	0.646	0.648	yes
Naive XGBoost	0.648	0.640	0.647	yes
Naive LightGBM	0.618	0.622	0.623	yes
Naive CatBoost	0.647	0.625	0.646	yes
Corrected AdaBoost	0.636	0.641	0.644	yes
Corrected GBM	0.648	0.646	0.648	yes
Corrected XGBoost	0.648	0.640	0.647	yes
Corrected LightGBM	0.618	0.622	0.623	yes
Corrected CatBoost	0.647	0.625	0.646	yes
RN AdaBoost	0.647	0.645	0.645	yes
RN GBM	0.647	0.645	0.645	yes
RN XGBoost	0.644	0.639	0.642	yes
RN LightGBM	0.647	0.647	0.641	yes
RN CatBoost	0.645	0.641	0.646	yes
CS AdaBoost	0.647	0.639	0.647	yes
CS GBM	0.633	0.634	0.633	yes
CS XGBoost	0.648	0.640	0.647	yes
CS LightGBM	0.621	0.625	0.619	yes
CS CatBoost	0.647	0.625	0.646	yes
AdaPU	0.645	0.627	0.642	yes
PU gradient boosting	0.636	0.648	0.649	yes

Two-sample - simulation: High dimension ($p = 500$)

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.563	0.576	0.587	yes
uPU logistic	0.526	0.533	0.539	yes
nnPU logistic	0.561	0.587	0.589	yes
Naive AdaBoost	0.615	0.643	0.642	yes
Naive GBM	0.615	0.638	0.641	yes
Naive XGBoost	0.620	0.641	0.643	yes
Naive LightGBM	0.603	0.640	0.638	yes
Naive CatBoost	0.614	0.637	0.638	yes
Corrected AdaBoost	0.615	0.643	0.642	yes
Corrected GBM	0.615	0.638	0.641	yes
Corrected XGBoost	0.620	0.641	0.643	yes
Corrected LightGBM	0.603	0.640	0.638	yes
Corrected CatBoost	0.614	0.637	0.638	yes
RN AdaBoost	0.620	0.645	0.646	yes
RN GBM	0.616	0.643	0.651	yes
RN XGBoost	0.619	0.657	0.644	yes
RN LightGBM	0.606	0.640	0.638	yes
RN CatBoost	0.617	0.656	0.657	yes
CS AdaBoost	0.615	0.645	0.641	yes
CS GBM	0.607	0.644	0.645	yes
CS XGBoost	0.617	0.646	0.653	yes

Two-sample - simulation: High dimension ($p = 500$)

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
CS LightGBM	0.607	0.642	0.640	yes
CS CatBoost	0.614	0.637	0.638	yes
AdaPU	0.625	0.640	0.642	yes
PU gradient boosting	0.618	0.647	0.648	yes

Two-sample - simulation: Curved boundary

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.622	0.642	0.642	yes
uPU logistic	0.620	0.641	0.641	yes
nnPU logistic	0.620	0.641	0.641	yes
Naive AdaBoost	0.614	0.644	0.643	yes
Naive GBM	0.615	0.642	0.642	yes
Naive XGBoost	0.614	0.635	0.645	yes
Naive LightGBM	0.599	0.628	0.631	yes
Naive CatBoost	0.623	0.639	0.638	yes
Corrected AdaBoost	0.614	0.644	0.643	yes
Corrected GBM	0.615	0.642	0.642	yes
Corrected XGBoost	0.614	0.635	0.645	yes
Corrected LightGBM	0.599	0.628	0.631	yes
Corrected CatBoost	0.623	0.639	0.638	yes
RN AdaBoost	0.625	0.628	0.636	yes
RN GBM	0.616	0.640	0.639	yes
RN XGBoost	0.614	0.635	0.641	yes
RN LightGBM	0.608	0.632	0.632	yes
RN CatBoost	0.622	0.642	0.643	yes
CS AdaBoost	0.615	0.644	0.643	yes
CS GBM	0.617	0.639	0.636	yes
CS XGBoost	0.616	0.644	0.642	yes
CS LightGBM	0.604	0.631	0.638	yes
CS CatBoost	0.623	0.639	0.638	yes
AdaPU	0.602	0.633	0.636	yes
PU gradient boosting	0.619	0.636	0.639	yes

Two-sample - simulation: Feature interaction

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.588	0.600	0.605	yes
uPU logistic	0.589	0.599	0.604	yes
nnPU logistic	0.589	0.599	0.604	yes
Naive AdaBoost	0.608	0.590	0.608	yes
Naive GBM	0.579	0.611	0.611	yes
Naive XGBoost	0.593	0.614	0.615	yes
Naive LightGBM	0.590	0.607	0.611	yes
Naive CatBoost	0.620	0.646	0.643	yes
Corrected AdaBoost	0.608	0.590	0.608	yes
Corrected GBM	0.579	0.611	0.611	yes
Corrected XGBoost	0.593	0.614	0.615	yes

Two-sample - simulation: Feature interaction

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Corrected LightGBM	0.590	0.607	0.611	yes
Corrected CatBoost	0.620	0.646	0.643	yes
RN AdaBoost	0.553	0.603	0.601	yes
RN GBM	0.558	0.599	0.598	yes
RN XGBoost	0.546	0.589	0.595	yes
RN LightGBM	0.543	0.594	0.584	yes
RN CatBoost	0.555	0.597	0.596	yes
CS AdaBoost	0.608	0.590	0.608	yes
CS GBM	0.579	0.611	0.611	yes
CS XGBoost	0.597	0.614	0.619	yes
CS LightGBM	0.590	0.607	0.611	yes
CS CatBoost	0.620	0.643	0.643	yes
AdaPU	0.582	0.580	0.599	yes
PU gradient boosting	0.579	0.588	0.606	yes

Two-sample - simulation: Unequal class covariance

Method	Same-model quantile	PU-estimated BA	Oracle BA	Verified
Logistic probability correction	0.646	0.655	0.657	yes
uPU logistic	0.646	0.657	0.657	yes
nnPU logistic	0.646	0.657	0.657	yes
Naive AdaBoost	0.633	0.641	0.658	yes
Naive GBM	0.634	0.657	0.658	yes
Naive XGBoost	0.635	0.659	0.652	yes
Naive LightGBM	0.621	0.641	0.640	yes
Naive CatBoost	0.639	0.661	0.665	yes
Corrected AdaBoost	0.633	0.641	0.658	yes
Corrected GBM	0.634	0.657	0.658	yes
Corrected XGBoost	0.635	0.659	0.652	yes
Corrected LightGBM	0.621	0.641	0.640	yes
Corrected CatBoost	0.639	0.661	0.665	yes
RN AdaBoost	0.613	0.632	0.639	yes
RN GBM	0.614	0.634	0.636	yes
RN XGBoost	0.603	0.622	0.634	yes
RN LightGBM	0.606	0.626	0.626	yes
RN CatBoost	0.616	0.639	0.641	yes
CS AdaBoost	0.633	0.641	0.658	yes
CS GBM	0.634	0.657	0.658	yes
CS XGBoost	0.638	0.653	0.654	yes
CS LightGBM	0.620	0.624	0.638	yes
CS CatBoost	0.639	0.661	0.665	yes
AdaPU	0.664	0.676	0.676	yes
PU gradient boosting	0.638	0.647	0.658	yes

Note. Balanced accuracy from one saved split/repetition per setting. – indicates an unverified replay, not a zero. Six real method/case replays are unverified; 1,194/1,200 method/case results are verified for each rule. Same-model quantile is the common prevalence quantile; oracle BA uses true validation labels only. PU-estimated BA remains a limited pilot.